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Assessing distributive inequities in FEMA's Disaster recovery assistance fund allocation

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ABSTRACT

Social vulnerability is significant in differential disaster impacts and outcomes and has been shown to influence the pace and progression of post-disaster recovery processes. Various disaster assistance programs - designed to support individual survivors, states, local governments, non-governmental entities, and businesses - may be distributing funds inequitably because they are not accounting for social vulnerabilities. One such program, the Federal Emergency Management Agency's Individuals and Households Program (IHP) program supports homeowners and renters for rental, repair, replacement, and permanent housing construction of disaster-damaged houses as well as providing other needs assistance to disaster victims with disaster-related non-housing expenses. To date, limited exploration connecting underlying social vulnerabilities and distribution of federal disaster assistance funds has provided only an incomplete understanding of the connections between underlying socio-economic conditions and often inequitable federal recovery assistance distribution. We analyzed county-level IHP assistance provided to homeowners between 2010 and 2018 and explored the relationships between fund distribution and social vulnerability indicators while controlling for total damages. Employing ordinary least squares regression analysis, we assessed links between Total Assistance, Housing Repair/Replace Assistance and Other Needs Assistance against social vulnerability variables from the Social Vulnerability Index (SoVI) framework. Results highlight specific social vulnerability variables influencing FEMA's distribution of disaster recovery assistance with race/ethnicity-related variables as the most frequent (regularly negative) influence on recovery fund receipt. Findings suggest that disaster assistance receipt is linked to underlying social vulnerability. As such, greater attention to and inclusion of social vulnerabilities would lead to more equitable distribution of disaster recovery funds.

1. Introduction

Disasters, by definition, are severe events that can cause extensive physical damage or financial losses to houses, critical infrastructure, transportation networks, agriculture, and businesses [1]. Major disasters, which exceed local response capacity, often force individuals, families, and communities to alter their everyday routines, relocate to safer areas, or seek assistance from outside their social networks and local communities. Large-scale and destructive disaster events are becoming more frequent and impactful in the

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United States [2]. In 2017 alone, ten Major Disasters impacted the United States and its Territories with three Hurricanes (Harvey, Irma, and Maria), causing an estimated $265 billion in damages and widespread displacement of survivors [3] and this trend has only increased over the past two-decades [4]. These disasters, increasing in frequency, magnitude, and per capita disaster losses, result in adverse outcomes for individuals and families [5,6]. Unfortunately, hazard losses are often inequitably borne by individuals and communities with less ability to prepare for, respond to, and rebound from disasters [7–9]. Identifying links between social vulnerability indicators, disaster impacts, and governmental support remains an important yet understudied topic. Although much is known conceptually about what makes individuals and communities more vulnerable to disasters, shocks, and stresses [10], relatively little is known about how these concepts manifest during the disaster recovery process - specifically in terms of the distribution of often scarce financial disaster recovery support.

Following disruptive disaster events, residents and communities often desire a rapid return to normalcy [11]. However, a disaster recovery's pace and trajectory are primarily determined by when and how requisite support is provided to disaster survivors [12]. Even with massive investment in research, development, and implementation of new recovery programs and policies, more abundant impact and recovery data, and technological advancement in computing capacity, society continues losing more to disasters [13,14]. Such losses are typically unequally borne by those most marginalized and vulnerable communities who struggle to prepare for, respond to, and rebound from disasters [15,16]. Previous research has shown that generally, socio-economic and demographic conditions, population density, pre-existing vulnerability, access to and receipt of aid packages, quality of governance, and damages from the disaster are primary determinants of post-disaster recovery progress [7,17–20].

Although many small scale disasters - impacting limited numbers of individuals and families - do not exceed the local capacity to respond and do not result in federal aid distribution, those that do (classified as Presidential Declared Disasters or Major Disasters) cause significant impact on families and communities (Fig. 1) [21]. In the United States, post-disaster recovery/rebuilding assistance to individuals and families impacted by major disasters is available from the Federal Emergency Management Agency's (FEMA) Individuals and Households Program (IHP) program only if the President grants such aid through a Presidential Disaster Declaration (PDD). Although other types of financial support are also available to disaster-affected households, including low-interest loans from Small Business Administration (SBA) which are uniquely tied to the IHP program, individual assistance (IA) grants for housing repair and replacement (HA) are directly given to disaster survivors with no repayment requirement. This paper explores the distribution of IHP Housing Assistance (HA) and Other Needs Assistance (ONA) provided by FEMA to individual disaster survivors and households after Major Disasters and their relationship with a suite of social vulnerability indicators. Specifically, this paper attempts to answer the following research questions: a) are social indicators of vulnerability associated with receipt of lower amounts of disaster recovery assistance when controlling for damage? b) which social vulnerability indicators are most influential in predicting IHP assistance? And c) do links between social indicators and recovery assistance funding indicate distributive inequities?

2. Background

2.1. Social equity

Environmental justice scholarship provides useful guidance for conceptualizing social equity where equity is linked to the notion of justice and implies fair treatment for all. According to preeminent environmental justice scholar Robert Bullard, social equity assesses "the role of sociological factors, such as race, ethnicity, class, culture, lifestyles, political power, and so on, in environmental decision making. In one example of social inequity, poor people and people of color often work in the most dangerous jobs and live in the most polluted neighborhoods, and their children are exposed to a host of environmental toxins on the playgrounds and in their homes" [22]. Social equity recognizes the historical, political, social, and economic contexts that structurally influence the prospects for access, opportunity, and outcomes. Socially vulnerable and marginalized populations deprived of resources to deal with acute and chronic shocks and stressors will continue experiencing adverse outcomes long after those who can access requisite resources. The distribution of social inequities is not randomly distributed; instead, they follow predictable social and geographical patterns linked to historical contexts where marginalized and vulnerable populations often receive inequitable treatment and undesirable outcomes [23]. Thus, allocation of resources should be prioritized based on the needs of individuals, households, and communities and should result in fair

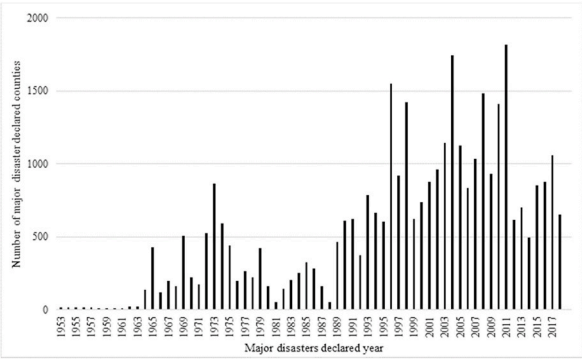

Fig. 1. Number of major disasters declared counties per year.

outcomes rather than simply treating everyone the same regardless of need when distributing goods and services [24,25].

In this article, we view social equity in terms of just allocation of resources across all social strata and adopt Rawlsian's distributive justice leans [26]. Rawl's theory of justice is essentially concerned with the role played by institutions in promoting justice. He argues, in the Difference Principle (DP), that inequalities can only be considered fair if they work to the benefit of the least advantaged members of the society, thereby mitigating inequalities of opportunities and the morally arbitrary effects of social and natural lotteries [26]. Rawl's distributive justice, in a disaster context, refers to the equitable distribution of society's risk, impact, outcome, and benefits. Disaster justice is then concerned with assisting where "the heaviest burdens of disasters being borne by those with the least power, - those who, for whatever social and economic reasons, are more exposed, more susceptible, and less resilient when disaster strikes" [27]. Although disasters result in negative outcomes for most individuals and communities, their impacts are especially impactful for the poor, weak, marginalized, and disadvantaged in society [28].

Inequities in the United States have been documented empirically across different policy domains [23], but such analysis is lacking in disaster recovery research [29,30]. In general, social equity measures are focused on assessing the extent to which disparities exist between different population groups. Equity analysis offers an assessment of the past based on the idea that what we learn from and improve our response to present and future disaster preparedness, response, and recovery efforts through evidence-based decision making. Empirical assessments of social equity are becoming more widespread with increasing data availability and more robust statistical tools. Results of such assessments can provide useful information to those local, state, and federal entities interested in evaluation and operational effectiveness in lessening social inequities [31].

Equitably distributing resources can significantly influence disaster recovery for the most marginalized in our society. Often, socially vulnerable populations, including minorities, poor, elderly, and female-headed households, receive a lower amount of aid [7,18,32,33]. African Americans and other minorities have been deliberately neglected by authorities after disasters [34], and some research argues that government agencies respond differently to the wealthy than to the poor [35]. Research on Hurricane Katrina recovery found that race matters in disaster recovery. Specifically, black workers from New Orleans were four times more likely than white counterparts to lose their jobs after the storm, all else equal [17]. Following severe flooding in Iowa in 2008, Hispanic and elderly populations were unable to secure affordable housing outside of the floodplain and experienced slower recovery rates as well as distributive and temporal inequalities in the allocation of federal recovery funds [36]. Thus, inequities in disaster recovery fund distribution require further investigation in order to address these challenges in the future.

2.2. Social vulnerability links to social equity

Social vulnerability is generally defined as a set of the pre-existing (individual, household, or community) conditions that hinder their disaster preparedness, risk mitigation, and recovery capacity and disproportionately increases disaster damage and loss burden [10,16]. Social vulnerability theories postulate that characteristics such as race and ethnicity, gender, income, poverty, age, education, and religion play significant roles individually or in combination (poor, minority and women) to exacerbate negative disaster

Table 1
Concept, description, and variable name for social vulnerability indicators used in this analysis.

Social Vulnerability Concept	Description	Variable Name (for use in models)
Employment Structure	Percent civilian unemployment	QCVLUN
	Percent employment in extractive industries	QEXTRCT
	Percent employment in service industry	QSERV
	Percent female participation in labor force	QFEMLBR
Housing	Percent renters	QRENTER
	Percent mobile homes	QMOHO
	Percent unoccupied housing units	QUNOCCHU
Population Structure	Percent population under 5 years or 65 and over	QAGEDEP
	Percent of children living in 2-parent families	QFAM
	Median age	MEDAGE
	Percent female	QFEMALE
	Percent female-headed households	QFHH
	People per unit	PPUNIT
Race/Ethnicity	Percent Asian	QASIAN
	Percent African American	QBLACK
	Percent Hispanic	QHISP
	Percent Native American	QNATAM
Socioeconomic Status	Percent poverty	QPOVTY
	Percent households earning over $200,000 annually	QRICH200K
	Per capita income	PERCAP
	Percent with less than 12th grade education	QED12LES
	Median housing value	MDHSEVAL
	Median gross rent	MDGRENT
	Percent of households spending more than 40% on housing costs	QHSEBURDEN
Special Needs	Percent households receiving social security benefits	QSSBEN
	Percent speaking English as a second language with limited English proficiency	QESL
	Nursing home residents per capita	QNRRES
	Percent of housing units with no car	QNOAUTO

*Variables removed due to collinearity are presented in supplementary materials (Table S2).

outcomes [37–39]. These same factors also influence where people live, their access to social and economic resources, and, therefore, their overall ability to cope and recover from disaster impacts [15]. While academia has highlighted the importance of the social vulnerability concepts and devised the use indicators in disaster preparedness, risk mitigation, and post-disaster recovery programs, local, state, and federal programs focused on disaster risk, reduction, mitigation, response, and recovery are only recently considering the concept of social effects [40–42].

For this study, we utilize computed data inputs to Cutter et al.'s (2003) social vulnerability index (SoVI) as indicators of vulnerability's influence on recovery outcome and to measure social equity. The SoVI index inputs provide a robust indicator set founded on a long history of disaster case studies, and it is the most widely used method to provide a societal context for measuring vulnerability to environmental hazards. The method has been applied in the U.S. and other country contexts and has found its way into many practical applications reaching beyond the academic spectrum, especially for preparedness and planning by disaster practitioners [43–46]. SoVI has evolved over time to include a comprehensive list of scalable, interrelated sociodemographic characteristics [47]. While the original method employed 42 variables, including some built environment indicators, the current SoVI model has been condensed to 28 variables representing six main vulnerability concepts: Employment Structure, Housing, Population Structure, Race/Ethnicity, Socioeconomic Status, and Special Needs (Table 1). SoVI input variables spatially pinpoint those populations generally characterized as having a lower capacity to prepare for, respond to, and rebound from disasters. Culled from decades of disaster case studies pinpointing the places and people who have seen adverse impacts from disasters, SoVI inputs represent a comprehensive suite of indicators that have proven useful for understanding diminished capacity in the face of disaster events. Because the disaster disciplines recognize the utility of social vulnerability variables as indicators of adverse outcomes, it stands to reason that they may also prove useful in understanding the distribution of often scarce post-disaster recovery program funds. Identifying links between pre-event social vulnerabilities and disaster outcomes is an integral step in assessing and correcting social inequities.

2.3. Disaster recovery and federal disaster assistance

Disaster recovery, defined as a process of restoring, rebuilding, and reshaping physical infrastructure, repopulation by survivors and new residents along with the gradual resumption of regular daily routines, ultimately returning to a stable state following a disaster [48,49], is both a process and a singular measurable outcome. As a process, disaster recovery is highly dependent on damages, resource availability, local-to-nation recovery program execution timelines, among other factors. No matter what shape recovery takes, it can broadly be divided into three phases: short, intermediate, and long term [50,51] (Fig. 2). Short-term recovery immediately follows a disaster event, is generally short-lived (days to weeks), and focuses on the distribution of relief materials such as food, water, and emergency shelter. Intermediate-term recovery generally focuses on the rehabilitation of disaster-impacted infrastructures, homes, and survivors. This phase, lasting weeks to months, aims to restore basic services. Long-term recovery, lasting months to years, includes a goal to reconstruct affected communities in a way that makes them more resilient to future disasters.

In the United States, disaster recovery is part of a larger framework of federal disaster relief provided by the Stafford Disaster Relief and Emergency Assistance Act, passed into law in 1988 [52]. When a disaster strikes a state, U.S. territory, or tribal area and exceeds local response and recovery capacity, the Governor or Tribal Chief Executive can formally request the President for a major disaster or emergency declaration [53]. Upon approval, a Presidential Disaster Declaration (PDD) will unlock federal response and recovery assistance to designated areas. Generally, PDDs include FEMA-supported disaster relief through two specific programs: a) the Public Assistance (PA) program, and b) the Individual Assistance (IA) Program. FEMA's PA program supports state, local and tribal governments to repair damage to public property and can include debris removal, emergency protective measures, and the restoration of disaster-damaged facilities [29]. FEMA's IA program provides support to families and individual disaster survivors for housing rental, repair and replacement assistance, and other needs assistance for disaster-related non-housing expenses such as replacing damaged household items, transportation, disaster-related medical and/or funeral expenses [54].

FEMA's IA program consists of six different types of programs: Mass Care And Emergency Assistance (MC/EA), Crisis Counseling Assistance And Training Programs (CCP), Disaster Unemployment Assistance (DUA), Disaster Legal Services (DLS), Disaster Case Management (DCM), and Individuals And Households Program (IHP) [53]. IHP provides financial assistance and direct services to eligible individuals and households who have uninsured or underinsured necessary expenses and serious needs, primarily contributing short to intermediate phases of disaster recovery (Fig. 2). FEMA IHP is not a substitute for insurance and cannot compensate for all losses caused by a disaster. Rather, it is intended to meet basic needs and supplement disaster recovery fields [53]. Currently, IHP grants may not exceed $36,000 (FY2021) per individual or household [55], although this amount is adjusted annually to reflect consumer price changes; and is generally limited to a period of 18 months [52]. IHP includes two categories of assistance, each with different scope and serving potentially diverse populations: Housing Assistance (HA) and Other Needs Assistance (ONA). Specifically, FEMA's HA program helps bridge the gap from sheltering/interim housing to permanent housing. The HA funds can be used for limited basic home repairs as well as rental payments for temporary housing. ONA supports other necessary expenses such as a funeral, medical, dental, childcare, critical needs assistance, clean and removal assistance, household items, moving and storage of personal items, transportation assistance, and group flood insurance policy.

IHP plays a crucial role in disaster recovery since timely receipt of post-disaster recovery assistance governs the trajectory of the entire recovery process [52]. This is especially true for areas with increased social vulnerability where limited resources often lead to a lack of ability to prepare, decreased ability to absorb losses, and lower recovery capacity. Previous studies indicate that access to federal disaster assistance is a key determinant of financial recovery after a disaster [56]. Based on an analysis of cash assistance following 32 tornadoes, Gallagher and colleagues found that disaster survivors who have access to federal assistance have $260-$1400 less in credit card debt than their counterparts. However, related research finds that disaster survivors in low-income and high minority communities encountered challenges in obtaining such assistance [57]. Our work attempts to understand the relationship

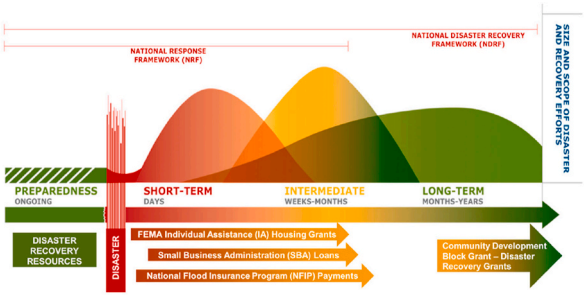

Fig. 2. Recovery continuum and federal recovery programs (adapted from Emrich et al., 2019).

between receipt of disaster funds and underlying socio-economic conditions across all of FEMA's IA declared counties for the contiguous United States (Fig. 3).

3. Materials and methods

3.1. Data

Data on Disaster Declarations (OpenFEMA Dataset: Disaster Declarations Summaries - v2) [21] and Individual and Housing Program (IHP) (OpenFEMA Dataset: Housing Assistance Program Data – Owners – v2) [58] were collected from the openFEMA database and social vulnerability variables, generated from U.S. Census 5-Year ACS data, were gathered from the University of Central Florida's Vulnerability Mapping and Analysis Platform [59]. Although IHP datasets contain information on both owners and renters, this study focuses on owners because renters are automatically ineligible for repair and replacement assistance - the largest component of total IHP assistance [60]. FEMA's *Owner* dataset contains data on FEMA's IHP program applicants, inspections, damage, and recovery assistance summarized at the county level for the conterminous United States (Fig. 2). Table 2 provides a list of variables from openFEMA datasets that are used in the current analysis.

Social vulnerability indicators from the Social Vulnerability Index (SoVI) [10] approach reflects a broad suite of socio-economic and demographic indicators shown to affect individual/community capacity to prepare for, respond to, and rebound from environmental shocks and stresses – like disasters. Here, twenty-eight variables related to six vulnerability concepts such as employment structure, housing, population structure, race/ethnicity, socio-economic status, special needs are included in the analysis. Because ACS 5-year data, which began in 2010, are composed fundamentally different from previous census data (pre-2010), this study analyzes Disaster Declaration and IHP data from 2010 to 2018 only.

3.2. Methods

Major Disasters or Presidential Declared Disasters (PDD) were curated from OpenFEMA Dataset: Disaster Declarations Summaries and were joined to the OpenFEMA Dataset: Housing Assistance Program Data – Owners – v2 dataset to create a composite county-level dataset of historical presidential disaster declarations, their impacts, and FEMA IHP support. Several *impacts* and *recovery* support variables were selected or created for analysis from these datasets, including: "valid registrations", "total inspected", "approved for FEMA assistance", "total damage", "total approved IHP amount", "repair/replace amount", and "other needs amount". Three standardized response variables (Table 3) were created by dividing the raw value for each county by the total approved for FEMA assistance in each county. Afterward, they were joined to social vulnerability variables (Table 1) using a unique combination of state and county that identify all administrative/legal and geographic areas for which Census Bureau tabulates data – more commonly known as GEOID or Federal Information Processing Standards (FIPS) codes. All social vulnerability indicators are normalized using percentage, density, and other functions following Cutter et al. (2003).

The Ordinary Least Squares (OLS) regression method was employed to identify primary relationships between social vulnerability variables and IHP assistance. Regression analysis, a statistical technique for investigating and modeling relationships between variables, determines the expected relationship between the response (impacts/damage and recovery assistance funds) and the social vulnerability predictor variables [61].

Social vulnerability variables correlating at the coefficient value of 0.8 (p-value < 0.5) or higher were removed prior to regression modeling for each model.[1] Finding that the response variables were highly skewed, a natural log transformation of the response variables was performed. Total damages were adjusted to $2018 using consumer price index (CPI) multipliers to account for annual inflation. Total damage per approved FEMA applicant (TOTALDMGPA) was set as a control variable (Table 3) to ensure that higher damage levels did not control model outputs. Next, the OLS model was tested for homoscedasticity. All values above $\pm$ 6 standard regression residuals (outliers) were iteratively removed from several regression model runs,[2] and a final backward elimination method OLS test was re-run. Three regression output models, namely Total IHP Assistance, Housing Repair/Replace Assistance, and Other

[1] Variables removed after multicollinearity test for each regression model are presented in supplementary materials (Table S1).
[2] An example of scatter plots used to determine outliers are presented in Supplementary materials (Fig S1).

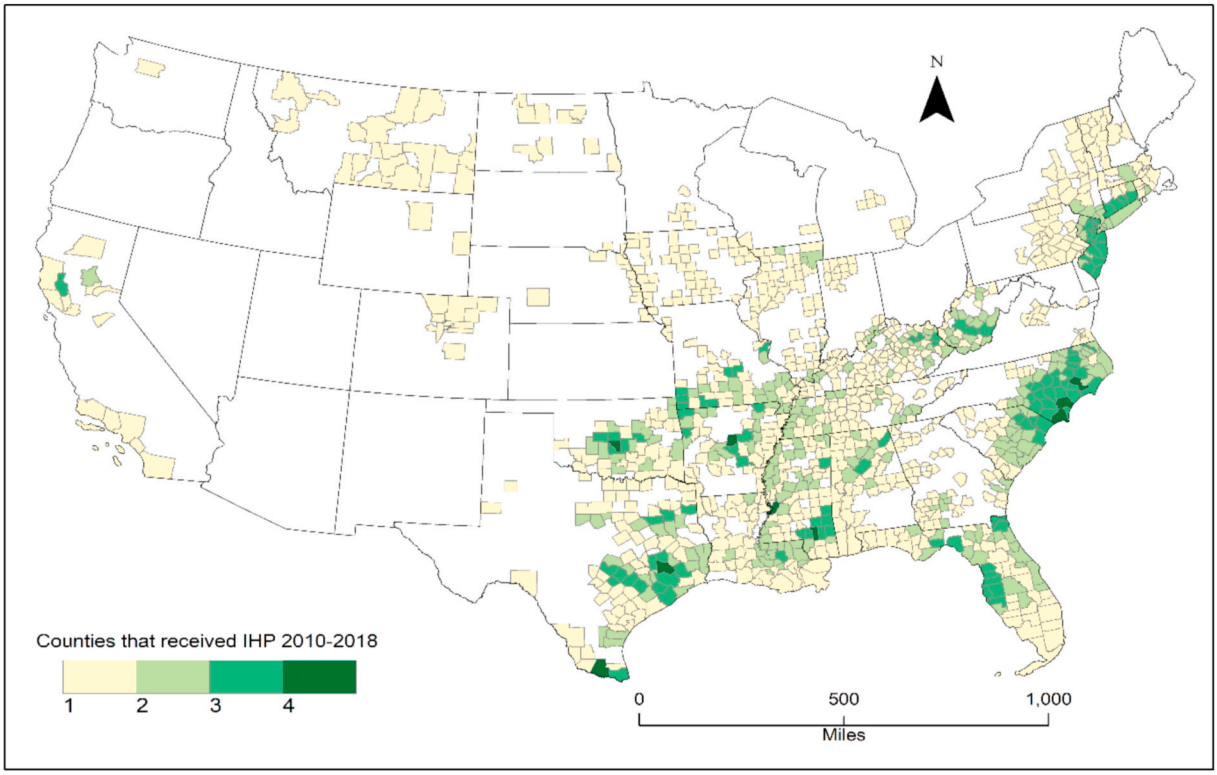

Fig. 3. Contiguous United States county study area (numbers reflecting the frequency of IHP assistance received by a county from 2010 to 2018).

Table 2
Selected data from Open FEMA Data used in this analysis.

Title	Description	Variable Name (for use in models)
Disaster Declarations data set (FEMA, 2020b)		
Disaster Number	Sequentially assigned number used to designate an event or incident declared as a disaster	disasterNumber
State	The name or phrase describing the U.S. state, district, or territory	state
Declared County Area	The name or phrase describing the U.S. County that was included in the declaration	declaredCountyArea
IH Program Declared	Denotes whether the Individuals and Households program was declared for this disaster.	ihProgramDeclared
Declaration Date	Date the disaster was declared	declarationDate
Disaster Type	Two-character code that defines if this is a major disaster, fire management or emergency declaration.	disasterType
Owner dataset (FEMA, 2020c)		
Disaster Number	Sequentially assigned number used to designate an event or incident declared as a disaster.	disasterNumber
State	Two letter abbreviations of the state (e.g., VA) where the disaster was declared	state
County	Full name of the County (e.g., Broward), followed by its type in parenthesis (e.g., county, parish, borough, etc.)	county
Valid Registrants	Count of FEMA registration owners within the state, county, zip where the registration is valid. In order to be a valid registration, the applicant must be in an Individual Assistance declared state and county and have registered within the FEMA designated registration period	validRegistrations
Total Damage	Total damage recorded by FEMA at the time of inspection	totalDamage
Approved for FEMA Assistance	Number of FEMA applicants who were approved for FEMA's IHP assistance	approvedForFemaAssistance
Total Approved IHP Assistance	Total amount approved in dollars under FEMA's IHP program	totalApprovedIhpAmount
Total Housing Repair or Replace Assistance	Total amount of Repair and/or Replacement approved in dollars for Housing Assistance (HA) under FEMA's IHP program (note that renters are not eligible for this type of assistance because they do not own the structure)	repairReplaceAmount
Total Other Needs Assistance (ONA)	Total amount of Other Needs (ONA) assistance approved in dollars under FEMA's IHP program (this could include, personal property, transportation, medical, dental, funeral, essential tools, moving/storage, miscellaneous and other needs)	otherNeedsAmount

Table 3
Response variable descriptions, treatments, and naming conventions used in this analysis. Data source: (FEMA 2020a, b).

Variable description	Treatment (variable X/variable Y)	Variable Name (for use in models	Model Name
Total approved IHP amount per approved for FEMA assistance	totalApprovedIhpAmount/ approvedForFemaAssistance	TOTALAPPROVEDPA	Total IHP Assistance model
Repair/replace amount per approved for FEMA assistance	repairReplaceAmount/ approvedForFemaAssistance	REPAIRREPLACEPA	Housing Repair/Replace Assistance model
Other needs amount per approved for FEMA assistance	otherNeedsAmount/ approvedForFemaAssistance	ONAPA	ONA model
Total damage per approved for FEMA assistance	totalDamage/approvedForFemaAssistance	TOTALDMGPA	Used as a control variable in all models

Needs Assistance (ONA) (Table 3), enabled assessment of interactions across all years (2010–2018) and for individual years meeting statistical requirements related to a number of inputs. Two years, 2013 and 2014, were dropped because of a significantly low number of total applicants which would have negatively influenced statistical power. All tests were run using SPSS version 25.0.

Since our study deals with many independent variables, we also ran an effect size and power analysis to gauge the effect and performance of our models. Effect sizes are an important complement to null hypothesis significance testing as they offer a measure of practical significance in terms of the magnitude of the effect, and are independent of sample size [62]. The effect size was estimated using Cohen's f^2 [63]. Effect size according to this test ranges from small ($f^2 = 0.02$), to medium ($f^2 = 0.15$), to large ($f^2 > 0.35$). In order to understand the power of the models, a power analysis was also performed primarily based on Cohen's work. Power is defined as the probability that a study will reject the null hypothesis when it is false. One of the common applications of statistical power analysis is to determine the power of study given the number of observations (N), size of the effect is the population (ES), and decision criterion used to determine statistical significance (α) [64]. Studies with high statistical power are very likely to detect the effects of treatments, interventions, and so on. To further examine the robustness of this finding, we conducted seemingly unrelated regressions (SUR) to check whether individual models are correlated to each other or not.

Assessing disaster recovery fund distribution from an equity perspective requires assessing each case where an influential relationship between SoVI variables and outcomes has been identified through Rawl's DP lens with the goal of determining if the relationship expresses (in)equity. For instance, Rawl's DP would call for more recovery funds going to people with less ability to prepare for, respond to, or rebound from disasters - socially vulnerable populations. Here, "least advantaged in society" are represented by the socially vulnerable. In other words, racial minorities, impoverished, less educated, renters, and others identified in the literature as having a lower capacity to prepare for, respond to, or rebound from disasters are considered "least advantaged." Classifying the model results in a measurable way to reflect Rawl's DP required us to summarize the number of times each social vulnerability indicator positively or negatively influenced receipt of FEMA IHP assistance across the three different sets of models. Displaying these summarized relationships (based on frequency and directionality) as either negative (inequitable), positive (equitable), or mixed equity provides a simple empirical measure across our resultant regression outputs. This classification scheme provided a visual depiction of directionality and frequency in the relationships between social vulnerability indicators and disaster outcomes. Based on this categorization, those variables leading to equitable funding distributions and those leading to less equitable distributions are more easily discerned. Recovery funding relationships contradicting this concept would be categorized as inequitable, while those showing that more funds are going to the most "needy" would be labeled equitable.

4. Results

The ordinary least square (OLS) tests identified influential social vulnerability predictors for three different outcome variables across all years (2010–2018) (Table 4) and annually for each outcome variable except Years 2013 and 2014 (Tables 5–7). We report only significant findings from these models, and variables are grouped into previously discussed vulnerability concepts (Table 1) for ease of interpretation. Our results indicated that all models have medium and large effect size (ES) and have desirable statistical power, as mentioned by Cohen (1988) [63]. All three All Year models have a large effect size, whereas a total of 18 Yearly models have a large and three Yearly models have a medium effect size (Table S31, S32, S33, and S34). Furthermore, except for the 2015 ONA Model, all models have enough power (Table S35, S36, S37, and S38). The detailed results can be found in the Supplementary materials (S3) section. Finally, seemingly unrelated regression (SUR) models yield similar results to individual OLS runs. Figs. 4–7 help explain patterns of inequity found by summarizing statistical results into counts for each social vulnerability determinant. Displaying these summarized relationships (based on frequency and directionality) as either negative (red shading showing negative relationships between predictor variables and outcome), positive (blue shading showing positive relationships between predictor and outcome), or mixed is a simple mathematical operation as seen in Figs. 4–7 enables easy identification of equity relationships within/across models. Each is discussed separately below.

4.1. All year model

Table 4 below shows the results of three All Year (2010–2018) models: Total IHP Assistance, Housing Repair/Replace Assistance,

Table 4
Influential social vulnerability predictors in Total IHP, Housing Repair/Replace, and ONA models across all years (2010–2018).

Concept	Variables	Total IHP Assistance model	Housing Repair/Replace Assistance model	ONA model
Race/Ethnicity	QBLACK	−0.134 (0.033)***	−0.277 (0.039)***	–
	QHISP	−0.17 (0.043)***	−0.387 (0.064)***	0.445 (0.085)***
	QNATAM	−0.407 (0.123)**	−0.439 (0.148)**	−0.539 (0.255)*
Special Needs	QNOAUTO	0.377 (0.118)**	0.588 (0.137)***	−0.839 (0.265)**
	QSSBEN	–	−0.413 (0.196)*	–
Population structure	QAGEDEP	−1.101 (0.166)***	−1.043 (0.326)**	–
	QFEMALE	–	–	1.304 (0.493)**
Housing	QUNOCCHU	0.181 (0.072)*	–	0.538 (0.148)***
	QRENTER	–	−0.377 (0.112)**	0.843 (0.197)***
	QMOHO	–	−0.255 (0.081)**	0.307 (0.137)*
Socioeconomic Status	QHSEBURDEN	−0.293 (0.08)***	–	−1.205 (0.19)***
	MDGRENT	–	0 (0)***	0 (0)***
	PERCAP	−0.000002629 (0)**	–	–
	QED12LES	–	0.343 (0.148)*	–
	QRICH200K	–	–	−0.207 (0.089)*
Employment Structure	QSERV	–	0.511 (0.187)**	–
	QEXTRCT	–	0.346 (0.14)*	–
	QCVLUN	–	–	1.008 (0.177)***
Control variable	TOTALDMGPA	0.00001463 (0)***	0.00001496 (0)***	0.00001591 (0)***
Model Descriptive	Constant	3.809 (0.05)***	3.91 (0.078)***	1.299 (0.319)***
	Adjusted R-square	0.482	0.471	0.269
	Number of applicants (sample size)	1,611	1,612	1,555

*** < 0.001, ** < 0.01, * < 0.05.
Standard errors are reported in parenthesis.

and ONA. Each column in this table represents final regression results from one of three models with significant variables, adjusted R-squared values, and the number of observations. For example, Total IHP Assistance model is composed of 11 significant variables combined in the following way[3]:

$$\begin{aligned} &log\ (TOTALAPPROVEDPA) \\ &\quad = 3.809(constant)\ -(0.134{*}QBLACK)\ -(0.17{*}QHISP)\ -(0.407{*}NATAM)\ +(0.377{*}QNOAUTO) \\ &\quad +(2.397{*}QNRRES)\ -(1.101{*}QAGEDEP)\ +(0.181{*}QUNOCCHU)\ -(0.293{*}QHSEBURDEN) \\ &\quad -(0.000002629{*}PERCAP) + (0.268{*}QSERV) + (0.00001463{*}TOTALDMGPA) \end{aligned}$$

Total IHP Assistance has a higher adjusted R-squared value than the other two models. The Total Funding model's adjusted R-squared value of 0.482 demonstrates that social indicators explain 48.2% of the variability of the data. Social vulnerability indicators explain 47.1% and 26.9% of the variability in Housing Repair/Replace assistance and ONA models, respectively. Three variables, Percent Hispanic, Percent Native American, and Percent without access to automobiles, and the control variable (Total Damage Per Approved Applicant) are significant across all three models, although the influence of these social indicators (positive/negative) varies by model. Race/Ethnicity variables QBLACK, QHISP, and QNATAM are negatively associated with total approved IHP per applicant and repair/replacement assistance per applicant. However, QNATAM has a negative influence in the ONA model, while QHISP has a positive relationship. Special Needs variable QNOAUTO is also significant in all models with a positive influence in Total IHP Assistance and Housing Repair/Replace Assistance models and negative influence in the ONA model. Likewise, QSSBEN is negatively influencing the Housing Repair/Replace Assistance model. Population Structure variable QAGEDEP negatively influences the outcome of Total IHP and Housing Repair/Replace models, while QFEMALE positively influences the ONA model. Housing variable QUNOCCUH positively influences Total IHP and ONA models, whereas QRENTER and QMOHO have a negative influence on the Housing Repair/Replace model and a positive influence on the ONA model. Socio-economic variable QHSEBURDEN negatively influences Total and ONA models. Furthermore, PERCAP and QRICH200K negatively influence Total and ONA models, respectively, and QED12LES has a positive influence on the Housing Repair/Replace model. Employment Structure variable QSERV and QEXTRCT positively influence receipt of Repair/Replace aid, and variable QCVLUN positively influences the receipt of ONA funds. As expected, the control variable TOTALDMGPA has a positive yet weak influence on all models.

Evaluating each regression equation's significant social vulnerability predictor variables from Rawl's DP perspective generates a different set of outcomes. Here, summarizing the frequency and direction of influence (positive or negative) of each significant social vulnerability on receipt of FEMA IHP funds provides a visual depiction of distributional (in)equities in FEMA's recovery programs (Fig. 4). The three "All Year" model outputs show that social vulnerability variables such as percent Native American, percent African American, percent housing burdened popuations, percent social security benefit recipients, and percent age-dependent population (under 5 and over 65 years) have systematically received lower amounts of Individual Assistance recovery support (total IHP, housing repair/replace, and ONA). Meanwhile, in these models, variables such as percent rich, per capita income, percent female population,

[3] All final regression models are presented in Supplementary materials (Table S2).

Table 5
Influential social vulnerability predictors in yearly Total IHP Assistance models (2010–2018)[a]. Each year column represents a different model output. Each model's descriptive stats (constant, number of applicants, adjusted R-square) are located at the bottom of each column.

Total IHP Assistance model								
Concept	Variables	2010	2011	2012	2015	2016	2017	2018
Race/Ethnicity	QBLACK	–	−0.19 (0.078)*	−0.237 (0.078)**	–	–	–	−0.555 (0.105)***
	QHISP	–	−0.531 (0.171)**	−0.725 (0.281)*	–	–	−0.537 (0.152)**	−0.264 (0.117)*
	QNATAM	−2.184 (0.54)***	−0.843 (0.206)***	–	−0.833 (0.242)**	–	–	–
Population Structure	QAGEDEP	−2.024 (0.439)***	2.356 (0.694)**	–	–	–	−2.081 (0.382)***	–
	MEDAGE	–	−0.013 (0.005)**	–	–	–	–	–
	QFEMALE	–	−1.121 (0.566)*	–	–	–	–	–
	QFHH	–	–	–	−2.071 (0.431)***	–	–	–
	QPUNIT	–	–	0.27 (0.076)**	–	0.105 (0.053)*	–	–
Special Needs	QNRRES	5.278 (1.949)**	–	–	–	–	–	–
	QNOAUTO	−0.996 (0.346)**	–	0.912 (0.192)***	–	1.684 (0.447)***	–	–
Housing	QMOHO	–	–	0.617 (0.168)***	–	–	−1.028 (0.205)***	–
	QRENTER	–	–	–	–	–	−0.981 (0.328)**	–
	QUNOCCHU	–	–	–	–	0.412 (0.13)**	–	–
Employment Structure	QCVLUN	0.555 (0.144)***	–		–	–	–	
	QSERV	–	–	–	–	–	–	1.826 (0.562)**
	QFEMLBR	–	–	−1.151 (0.532)*	–	–	–	–
Socioeconomic Status	QRICH200K	−1.868 (0.466)***	–	−0.967 (0.455)*	–	–	–	–
	QHSEBURDEN	–	–	1.482 (0.245)***	–	−1.365 (0.262)***	–	–
	QPOVTY	–	0.66 (0.221)**	−0.7 (0.282)*	–	–	−1.129 (0.519)*	–
	QED12LES	–	–	–	–	–	1.589 (0.554)*	–
	MDGRENT	–	–	–	–	0 (0)**	0 (0)**	–
Control Variable	TOTALDMGPA	0.00001763 (0)***	0.000008481 (0)***	0.00001358 (0)***	0.000002013 (0)***	0.00003218 (0)***	0.00001357 (0)***	0.000004145 (0)***
Model Descriptive	Constant	3.869 (0.108)***	4.135 (0.323)***	2.932 (0.309)***	3.912 (0.062)***	3.025 (0.164)***	4.697 (0.189)***	2.638 (0.388)***
	Adjusted R-square	0.532	0.475	0.708	0.257	0.684	0.601	0.619
	Number of applicants (sample size)	239	470	174	145	244	161	105

*** < 0.001, ** < 0.01, * < 0.05.
Standard errors are reported in parenthesis.
[a] 2013 and 2014 models did not meet the minimum number of observations (n) to run OLS confidently.

Table 6
Influential social vulnerability predictors in yearly FEMA IA housing repair/replace assistance models (2010–2018)[a]. Each year column represents a different model output. Each model's descriptive stats (constant, number of applicants, adjusted R-square) are located at the bottom of each column.

Housing Repair/Replace Assistance Models								
Concept	Variables	2010	2011	2012	2015	2016	2017	2018
Race/Ethnicity	QBLACK	–	−0.34 (0.075)***	−0.504 (0.103)***	–	–	–	−0.522 (0.117)***
	QHISP	–	−0.734 (0.172)***	−1.16 (0.393)**	–	–	−0.879 (0.167)***	−0.256 (0.128)*
	QNATAM	−2.222 (0.585)***	−0.94 (0.203)***	–	−0.683 (0.273)*	–	–	–
Population Structure	QAGEDEP	−3.4 (0.625)***	–		–	–	−3.106 (0.569)***	–
	QFHH	–	–	–	−2.652 (0.486)***	–	–	–
	QPUNIT	–	–	0.318 (0.113)**	–	–	–	–
Special Needs	QNOAUTO	–	–	–	–	1.727 (0.47)***	–	–
	QSSBEN	0.869 (0.277)**	–	–	–	–	–	–
	QNRRES	7.529 (2.167)**	–	–	–	–	–	–
Housing	QRENTER	–	–	0.883 (0.359)*	–	–	−1.36 (0.458)**	–
	QMOHO	–	−0.297 (0.143)*	0.582 (0.22)**	–	–	−1.786 (0.281)***	–
Employment Structure	QEXTRCT	−0.819 (0.339)*	0.536 (0.206)*	–	–	–	–	–
	QSERV	–	0.943 (0.33)**	–	–	–	1.747 (0.797)*	2.105 (0.709)**
	QFEMLBR	–	–	−1.371 (0.666)*	–	–	–	–
Socio-economic Status	QRICH200K	−1.819 (0.543)**	–	−1.421 (0.572)*	–	–	–	–
	QPOVTY	–	0.971 (0.24)***	−0.848 (0.411)*	–	−0.448 (0.218)*	–	–
	QHSEBURDEN	−0.697 (0.223)**	–	1.544 (0.306)***	–	−0.659 (0.179)***	−2.196 (0.477)***	–
	QED12LES	–	–	–	–	–	2.475 (0.558)***	–
Control Variable	TOTALDMGPA	0.00001736 (0)***	0.00001138 (0)***	0.00001403 (0)***	0.000002224 (0)***	0.00004334 (0)***	0.00001878 (0)***	0.000005166 (0)***
Model Descriptive	Constant	4.081 (0.141)***	3.261 (0.058)***	2.647 (0.47)***	3.861 (0.07)***	3.323 (0.057)***	4.548 (0.221)***	3.472 (0.153)***
	Adjusted R-square	0.491	0.484	0.669	0.262	0.718	0.622	0.607
	Number of applicants (sample size)	239	461	174	145	241	161	105

*** < 0.001, ** < 0.01, * < 0.05.
Standard errors are reported in parenthesis.
[a] 2013 and 2014 models did not meet the minimum number of observations (n) to run OLS confidently.

Table 7
Influential social vulnerability predictors in yearly FEMA IA Other Needs Assistance (ONA) models (2010–2018)[a,b]. Each year column represents a different model output. Each model's descriptive stats (constant, number of applicants, adjusted R-square) are located at the bottom of each column.

Other Needs Assistance (ONA) Models							
Concept	Variables	2010	2011	2012	2016	2017	2018
Race/Ethnicity	QBLACK	0.527 (0.215)*	–	–	–	–	−0.504 (0.192)*
	QHISP	0.62 (0.162)***	–	−1.22 (0.565)*	–	–	−0.592 (0.242)*
	QNATAM	−3.199 (1.233)*	–	–	–	−7.996 (3.071)*	–
Population Structure	QAGEDEP	−4.515 (1.041)***	–	–	–	–	–
	MEDAGE	–	−0.035 (0.008)***	–	–	–	–
	QPUNIT	–	–	–	0.307 (0.111)**	–	–
	QFAM	–	–	1.123 (0.325)**	–	–	–
Special Needs	QNRRES	9.124 (4.537)*	–	−19.682 (9.028)*	–	–	−21.877 (10.072)*
	QSSBEN	–	1.49 (0.552)**	–	–	–	–
	QNOAUTO	–	–	–	1.623 (0.69)*	–	–
Housing	QUNOCCHU	–	0.732 (0.259)**	–	–	–	–
	QMOHO	–	–	1.012 (0.323)**	–	–	–
	QRENTER	–	–	1.364 (0.426)**	–	–	–
Employment Structure	QSERV	–	−1.216 (0.558)*	–	–	–	2.189 (0.834)*
Socioeconomic Status	QHSEBURDEN	−1.211 (0.495)*	–	2.036 (0.497)***	−0.891 (0.338)**	–	–
	QRICH200K	−2.43 (1.101)*	–	–	–	–	–
	QED12LES	–	1.138 (0.395)**	–	–	1.554 (0.375)***	1.808 (0.664)**
	QPOVTY	–	–	–	–	−1.101 (0.411)**	–
Control Variable	TOTALDMGPA	0.0000238 (0)***	0.00001269 (0)***	0.00001322 (0)***	0.00002784 (0)***	0.000008023 (0)***	0.000003386 (0)***
Model Descriptive	Constant	3.336 (0.287)***	3.631 (0.521)***	–	1.747 (0.326)***	2.607 (0.062)***	2.026 (0.156)***
	Adjusted R-square	0.343	0.371	0.368	0.271	0.198	0.292
	Number of Applicants (sample size)	233	433	172	234	151	100

*** < 0.001, ** < 0.01, * < 0.05.
Standard errors are reported in parenthesis.
[a] 2013 and 2014 models did not meet the minimum number of observations (n) to run OLS confidently.
[b] 2015 model lacks desirable explanatory power. Details about power analysis and effect size are in Supplementary materials (S3).

percent less than 12th grade education, percent service sector employment, percent unemployment, and total damage per applicant have equitably received federal individual assistance. Several variables, including percent Hispanic population, percent mobile homes, percent renters, and percent without automobiles have depicted mixed equity across the models (Fig. 4). In these instances, at least one model depicts inequitable distribution of recovery resources while other show equitable distribution when controlling for total damage per applicant.

4.2. Individual year models

Gaining a more nuanced understanding of the interaction of social vulnerability indicators and disaster outcomes required individual model runs for each year of data. Because each year's model run used a different set of counties experiencing disasters with different magnitudes, this annual approach yielded a somewhat different set of disaster recovery linked social indicators for each year's model run. Statistical results were then summarized and assessed based on our definition of distributive justice to determine directionality (equitable, inequitable, or mixed equity) and frequency for each statistically social vulnerability variable.

4.2.1. Predictors of total IHP assistance model by year

Assessing linkages between social indicators and disaster assistance receipt at the county level while controlling for damages enables us to portray a broad picture of potential equity issues in disaster recovery assistance allocations. Here, OLS regression identified between 3 and 10 social vulnerability indicators in each model either positively or negatively linked with funding receipt.

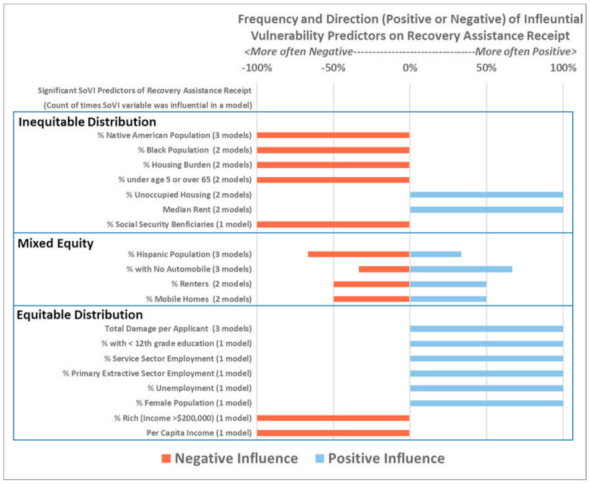

Fig. 4. Directionality and frequency of significant variable influence on receipt of FEMA IA total IHP funds, Repair/Replace funds, and ONA funds (2010–2018). Red shading shows a negative influence on IHP assistance receipt and blue shading shows positive influence. Percentages indicate frequency of significant influence (e.g. % Native American population was negatively influential 100% of the time it was found to be a significant predictor of total IHP assistance receipt (n = 3)). Equity categorization is based on a-priori application of Rawls' "Difference Principle" where one recognizes that the "least advantaged in society" – the socially vulnerable – should receive more recovery funds (controlling for damage) than those less vulnerable in order to affect positive change for the least advantaged. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

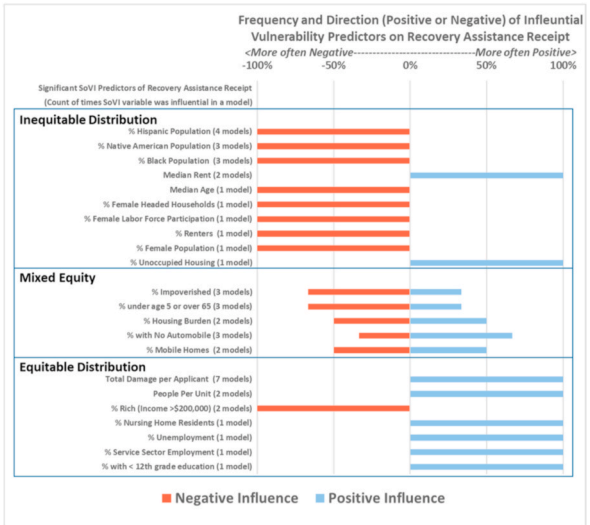

Fig. 5. Directionality and frequency of significant variable influence on receipt of FEMA IA total IHP funds (2010–2018 excluding 2013–14). Red shading shows negative influence on IHP assistance receipt and blue shading shows positive influence. Percentages indicate frequency of significant influence (e.g. % Hispanic population was negatively influential 100% of the time it was found to be a significant predictor of total IHP assistance receipt (n = 4)). Equity categorization is based on a-priori application of Rawls' "Difference Principle" where one recognizes that the "least advantaged in society" – the socially vulnerable – should receive more recovery funds (controlling for damage) than those less vulnerable in order to affect positive change for the least advantaged. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 5 presents only significantly influential social vulnerability predictors of total approved IHP amount per FEMA applicant approved for assistance (TOTALAPPROVEDPA) for each of seven years (models) assessed. The adjusted R-square value for these models of total IHP funding by year ranges from 0.257 to 0.708. All variables representing the Race/Ethnicity social vulnerability pillar show a significant negative influence on the amount of IHP received. QBLACK is influential in 2011, 2012, and 2018; QHISP is significant in 2011, 2012, 2017, and 2018; and QNATAM is negatively influential in 2010, 2011, and 2015. Variables associated with Population Structure have both positive and negative influences on receipt of total IHP assistance. While QPUNIT positively influences total IHP assistance receipt in 2012 and 2016, MEDAGE and QFEMALE are negatively influential for 2011. QAGEDEP is negatively related to total IHP in the 2010 and 2017 models but positively influential in the 2011 model. Also, QFHH is negatively influential in the 2015 model. Two Special Needs variables significantly influence receipt of IHP funding: QNRRES is positively influential for 2010, and QNOAUTO shows a positive influence in 2012 and 2016 but a negative influence for the 2010 model. Three Housing-related

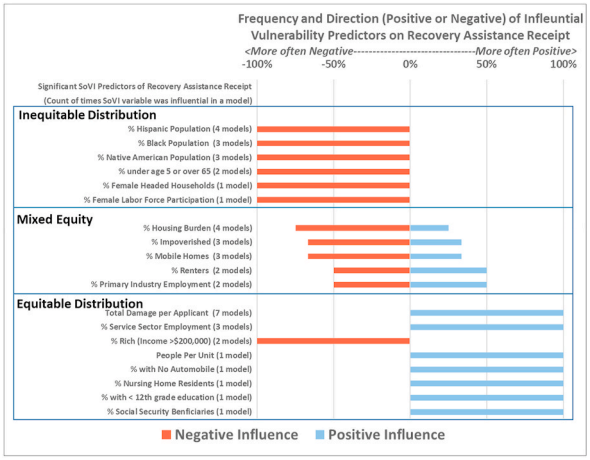

Fig. 6. Directionality and frequency of significant variable influence on receipt of FEMA IA Housing Repair and Replace Assistance funds (2010–2018 excluding 2013–14). Red shading shows a negative influence on housing repair/replace assistance receipt and blue shading shows a positive influence. Percentages indicate frequency of significant influence (e.g. % Hispanic population was negatively influential 100% of the time it was a significant predictor of total repair/replace assistance receipt (n = 4)). Equity categorization is based on a-priori application of Rawls' "Difference Principle" where one recognizes that the "least advantaged in society" – the socially vulnerable – should receive more recovery funds (controlling for damage) than those less vulnerable in order to affect positive change for the least advantaged. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

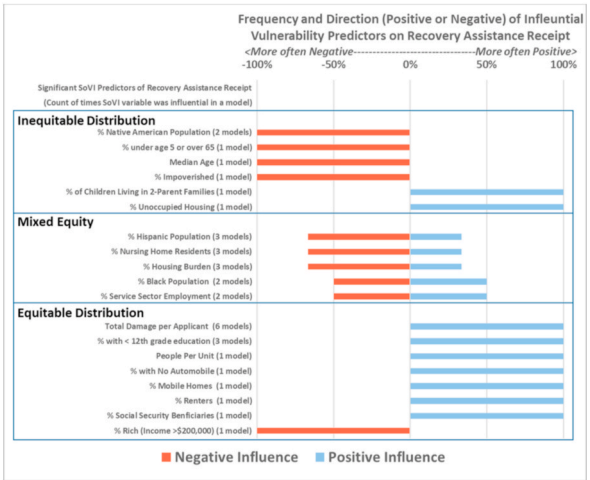

Fig. 7. Directionality and frequency of significant variable influence on receipt of FEMA IA Other Needs Assistance (ONA) funds (2010–2018 excluding 2013–14). Red shading shows negative influence on housing ONA receipt and blue shading shows positive influence. Percentages indicate frequency of significant influence (e.g. % Native American population was negatively influential 100% of the time it was found to be a significant predictor of total repair/replace assistance receipt (n = 2)). Equity categorization is based on a-priori application of Rawls' "Difference Principle" where one recognizes that the "least advantaged in society" – the socially vulnerable – should receive more recovery funds (controlling for damage) than those less vulnerable in order to affect positive change for the least advantaged. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

variables are significant in three yearly (2012, 2016, and 2017) models with percent mobile home positively influencing receipt in the 2010 model and showing a negative influence in the 2017 model along with percent renter populations. Likewise, three Employment Structure related variables are significant in different years (2010, 2012, 2018), with unemployment and service sector employment positively associated with IHP receipt and percent female labor force participation negatively associated with receipt of IHP assistance. Five different Socio-economic Status variables are significant in different years: QRICH200K is negatively influential in 2010 and 2012; QHSEBURDEN is positively influential in 2012 and negatively influential in 2016; QPOVTY is positively influential in 2011 but negatively influential in 2012 and 2018; QED12LES is positively influential in the 2017-year model. As expected, the control variable (damage per approved applicant) is highly significant and positively influential in each of the years assessed in the Total IHP Assistance model.

Assessing the same data from a Rawlsian equity perspective yields a different set of outcomes. Here, equitable distribution of recovery funds is generally linked with more people per unit, higher unemployment, lower educational attainment, higher % of service sector employment, and total damage per applicant (Fig. 5). Conversely, inequitable distribution of these disaster recovery funds – seen

in the top portion of Fig. 5 is influenced by a county's percentage of racial and ethnic minorities, % older populations, % renters, and % impoverished (among others). A sizable number of SoVI indicators show clear distributive inequities when controlling for damage. Here, all racial or ethnic vulnerability indicators found to be significant predictors of IHP were negatively influential 100% of the time. A third category, mixed equity, shows some years where these vulnerable groups saw equitable distributions and other years where their vulnerabilities were linked with inequitable receipt of disaster recovery funds.

4.2.2. Predictors of Housing Repair and Replace Assistance by year

Predictors of FEMA's Other Needs Assistance (ONA) Table 6 reports significant social vulnerability predictors of the Housing Repair/Replace Assistance model, which used repair/replace amount per approved applicant for FEMA assistance (REPAIRREPLACEPA) as an outcome variable. Model's adjusted R-squared values range from the lowest 0.262 to the highest 0.718. Three Race/Ethnicity related variables are negatively influential whenever they are significant in a model: QBLACK is negatively influential in 2011, 2012, and 2018; QHISP is negatively influential in 2011, 2012, 2017, and 2018; QNATAM is negatively influential in 2010, 2011 and 2015. Three Population Structure related variables are significant in different yearly runs of the Houring Repair/Replace Assistance model: QAGEDEP is negatively influential in 2010 and 2017; QFHH is negatively influential in 2015, and QPUNIT is positively influential in the 2012 yearly model. Three Special Needs variables are positively and significantly associated with housing repair/replace assistance in various years: percent housing units without a car is positively influential in 2016; percent households receiving social security benefits and nursing home residents per capita are positively influential in 2010. Two housing-related variables are significant in several model runs. Percent renters is positively influential in 2012 and negatively influential in 2017. Percent mobile homes is positively influential in 2012 and negatively influential in 2011 and 2017. Three variables related to Employment Structure are significant in yearly receipt of repair/replace assistance: QEXTRCT is negatively influential in 2010 and positively influential in 2011; QSERV is positively influential in all three years in 2011, 2017, and 2018; and QFEMLBR is negatively influential in 2012. Four Socio-economic Status related variables are significant in the yearly Repair/Replace Funding model: QRICK200K is negatively influential in 2010 and 2012; QPOVTY is negatively influential in 2012 and 2016, and positively influential in 2011; QHSEBURDEN is negatively influential in 2010, 2016, and 2017 and positively influential in 2012; and QED12LES is positively influential in 2017. The control variable, total damage per approved applicant, is highly significant and positively influential in all years of Housing Repair/Replace model.

Observing these model outputs from an equity perspective results in a comparable set of findings as those described above. Here again, links between racial and ethnic populations and lower levels of fund receipt become clear (Fig. 6). Percent Black, % Hispanic, and % Native American, along with % Age Dependent Populations, and Gender indicators of social vulnerability (Female Headed Households and % Female Labor Force Participation) all indicate higher social vulnerability or in terms of Rawlsian equity needy populations. Similarly, summarized results for FEMA Housing Repair/Replace Assistance show both mixed equity and equitable funding distribution. In terms of mixed equity, several social vulnerability factors lead to increased and decreased assistance in different models with no discernible overall directionality in equity. Percent renters, percent mobile homes, percent employed in primary industry, percent impoverished, and percent spending more than 30% of income on housing costs do not show a clear relationship (positive or negative) in receipt in repair/replace assistance. Interestingly, a higher total number of social vulnerability variables point toward equitable housing assistance grant support. Here, places with more people per household, lower auto access, higher percentages of nursing home residents, lower educational attainment, greater rates of service sector employment, and higher percentages of social security beneficiaries, and higher overall losses are receiving higher amounts of FEMA Housing Repair/Replace Assistance.

4.2.3. Predictors of other needs assistance (ONA) by year

Table 7 presents the influential social vulnerability variables predicting receipt of ONA aid in the yearly ONA model. The adjusted R-squared values for the ONA yearly model range from the lowest 0.198 to the highest 0.371. Three Race/Ethnicity variables are significant for different years: QBLACK is positively influential in 2010 and negatively influential in 2018; QHISP is positively influential in 2010 and negatively influential in 2012 and 2018; QNATAM is negatively influential in the 2010 and 2017 yearly model. Four different Population Structure variables are significant in this model: variables such as population under 5 years and 65 years, and median age are negatively influential, whereas the percent of children living in 2-parent families and people per unit are positively influential. Three Special Needs variables, QNRRES, QSSBEN, and QNOAUTO, are significant in the yearly ONA model: QNRRES is negatively influential in 2012 and 2018 and positively influential in 2011; QSSBEN and QNOAUTO are positively influential in 2011 and 2016 accordingly. Three Housing variables are positively influential and significant in different years: QUNOCCUH (2011) and QMOHO and QRENTER (2012). Variable QSERV, related to Employment Structure, is significant but negatively influential in 2011 and positively influential in 2018. Four variables related to Socioeconomic Status are significant in the ONA models: QHSEBURDEN is negatively influential in 2010 and 2016 and positively influential in 2012; QRICH200K is negatively influential in 2010; QED12LES is positively influential in years 2011, 2017, and 2018; and QPOVTY is negatively influential in 2017. The control variable TOTALDMGPA is positively influential and highly significant in all six years of the ONA model.

Assessing distributions from a Rawlsian equity approach produces similar results as both the total IHP and the housing repair/replace assistance model with several notable exceptions (Fig. 7). First, unlike the IHP and repair/replace models, % Black population and % Hispanic population resulted in mixed equity outcomes - in some cases positively influencing receipt and, in some cases, negatively influencing receipt. However, like the other models, % Native American remains a negative influence on receipt of ONA funds when it is significantly predictive. Further, population structure characteristics such as female-headed households, median age, and poverty influence inequitable receipt in a negative direction, while the percent of children living in 2-parent families positively

influences ONA receipt. From the Rawlsian perspective, these results indicate inequitable distribution because applying Rawl's DP here should lead to more recovery funds going towards places with lower percentages of children in two-parent families (higher percentages of children in single-parent families). Conversely, several relationships identified in this model represent the equitable distribution of disaster recovery funds, including higher funding going to homes with more people and places with higher percentages of families without access to automobiles, mobile homes, renters, low educational attainment, and social security beneficiaries. Unfortunately, in most of these cases, the number of times these equitable distributions occurred was low (n = 1) except in the case of social security beneficiaries (n = 7) and low educational attainment (n = 3).

5. Discussion

Results illustrate the utility of social vulnerability indicators as predictors of federal disaster recovery aid and highlight both positive and negative equity outcomes in FEMA's IHP program. Total damage per capita was the single most influential variable across all models indicating that FEMA's IHP program is serving the areas most impacted by disasters. Since IHP assistance is tied to damages and intended to make a home inhabitable again, the result is not only expected but evidence that FEMA's IHP recovery programs are operating (at least on the surface) as intended - where counties with the highest disaster damages are receiving the most recovery funding. Beyond this fact, when controlling for damages and undertaking a retrospective analysis of the relationships between losses and social vulnerabilities, some findings point toward unintended distributive inequalities in FEMA's IHP program that warrant further investigation and discussion. While it is not unexpected to find relationships between underlying social vulnerability indicators and receipt of FEMA IHP assistance, any negative relationships - those indicting inequitable distribution of recovery assistance - deserve additional inspection and perhaps a deeper look at how program objectives, guidance, policies, and administration may be influencing equitable recovery for all disaster survivors.

This analysis found that although various social vulnerability indicators were significant in different regression models, no specific social vulnerability indicators were consistently significant in all models. Among three outcome indicators assessed in the "All Year" the same variables were generally influencing assistance distribution, and near-identical relationships were observed in total IHP assistance and repair/replace assistance models. In contrast, the distribution of ONA model has a different set of significant variables and, most notably, the opposite relationship between variables and outcome than the other two models. This finding suggests that the dynamic interplay between underlying vulnerability characteristics and the disaster itself produces distinct outcomes that are not easily predicted by the univariate set of social characteristics so often used to explain disaster losses such as wealth/poverty or white/non-white.

The composite models (Total IHP, Housing Repair/Replace, and ONA) spanning the entire time period highlight that race/ethnicity variables (QBLACK, QHISP, and QNATAM) are linked with receipt of less total IHP assistance and lower housing repair/replace assistance funding. Likewise, the age-dependent populations are receiving less IHP assistance and repair/replace assistance funding, whereas the percentage of housing units without automobile access is linked with high receipt of both IHP and repair/replace funds. Similarly, lower amounts of ONA funds per capita were found in counties with higher percentages of Native Americans, higher percentages of housing units without a car, lack of wealthy residents (those earning more than $200,000 annually), and a higher percentage of households spending more than 40% in housing costs. Conversely, higher amounts of ONA funds are found in counties with a higher percentage of Hispanics, females, unoccupied housing units, renters, mobile homes, and civilian unemployment.

Assessing these relationships year-by-year for each different recovery funding pool again identified three race/ethnicity indicators of disaster assistance funds in all three outcome models. Specifically, results find that lower amounts of total IHP assistance are consistently provided to areas with higher percentages of African Americans, Hispanics, and Native Americans (Table 5). Although several other social vulnerability indicators are significant in several models, no consistent influence was found. A similar effect is observed in the distribution of repair/replace assistance (Table 6); that is, as percentages of racial and ethnic minority populations increase, the amount of repair/replace assistance decreases. Likewise, counties with higher QAGEDEP values have received lower amounts of repair/replace assistance funds. The model of distribution of ONA funds (Table 7) and its relationship with social vulnerability indicators does not correspond well to the other models. However, recent years show an increase in the negative relationship between race/ethnicity variables and receipt of ONA funds.

Our findings echo several previous works highlighting inequities in the delivery of disaster assistance. Various social vulnerability indicators, especially race/ethnicity-related variables, continue playing an important role in allocating federal disaster aid. Previous studies show that the distribution of FEMA individual assistance decreases with the increased presence of the Black and Asian populations [7,33]. Like these, we find that lower amounts of receipt of aid are also related (in some models) to socio-economic factors such as median income, age-dependent population, and renters [18,30,57].

6. Conclusion

The objectives of this study were to 1. identify how well leading social vulnerability variables were able to predict FEMA IHP assistance in disaster-affected counties between 2010 and 2018; 2. determine which indicators were most influential in predicting recovery fund allocation, and 3. quantify existing inequities in fund distribution from a Rawlsian Difference Principle perspective. County-level aggregated data obtained from the OpenFEMA database was used for the analysis and found that various social vulnerability variables are predictive of federal disaster aid allocation following presidential disaster declarations. Results further indicated that IHP recovery funds are often distributed inequitably when assessing these relationships at the county scale and accounting for total damages.

Although we explored the relationship of social vulnerability variables in receiving IHP assistance, we recognize that several

processes are involved in making the decisions on the disbursement of assistance that significantly impacts the final amount of aid. Registration of the disaster survivors and verification of damages are vital steps in determining that final amount of aid. Registration to the program is the foremost requirement, and the process is impacted by various social vulnerability factors such as employment status, vehicle ownership, language constraints, awareness about the disaster assistance programs, and application process [54,65. Because our models show that disaster damage is the most significant predictor in each model, it is important to understand how the damage assessment process is performed and if inequities are present from the first FEMA/survivor interactions. Previous research has documented some important relationships between social vulnerability variables such as foreign-born population and educational attainment influence damage assessments and assistance levels [60].

Our findings can help to improve how FEMA recovery programs operate. Since the frequency and severity of disasters are increasing, the need and demand for federal disaster assistance would also be increasing. Data on distributive inequities in disaster recovery fund allocation can provide FEMA and other concerned authorities with the information needed to modify programs, policies and implementation plans. Further, because disaster outcomes are the results of a complex interaction between hazardous processes and societal vulnerability, these relationships are not always easily explained by univariate social measures or compound vulnerability indicators. Key vulnerability variables, drawn from the literature, do not always perform as expected when assessing disaster-specific outcomes.

Major disasters significainntly disrupt the normal functioning of individuals, families, and communities, thus requiring external support to bounce back and move forward. Disaster recovery, both short-term and long-term, is uneven depending on the pre-existing conditions of disaster survivors, such as race, class, and gender. This research investigated the distribution of federal disaster assistance – IHP after a declaration of Major Disaster from 2010 to 2018 and its relationship with social vulnerability indicators at the county level. Individual assistance is a primary vehicle to support individuals and families in the aftermath of a disaster and an important resource for the disaster recovery process. The size of the United States, the complexity of intergovernmental relationships, the socio-political and geographic differences across regions, the increasing wealth gap, changes in frequency and severity of hazard events, and other complicating factors such as the wide range of capabilities and capacities between and across communities all serve to make disaster recovery programs implementation a challenge at any time. These discrepancies, spread out geographically, render a one-size approach to disaster recovery impossible. Hence, disaster recovery programs should be cognizant of findings such as those presented here to think critically about disaster recovery program development and implementation. We believe this research would help FEMA and other entities design more just and equitable disaster assistance distribution programs and policies. Improving on the old adage "if it is measured, it can be managed" by using social data to understand adverse disaster outcomes in a way that enables improvement in program delivery and implementation. However, socio-economic and demographic conditions of disaster survivors do not guarantee the even distribution of resources to all those involved. Finer resolitions (applicant level) assessments would illuminate these connections between social conditions and adverse outcomes. However, data to support these types of individual level analyses are not currently available for consumption and assessment. Although our study does not find a set of specific variables that influence individual assistance distribution, the negative association of race/ethnicity-related variables across the model supports broader findings of the research that disaster survivors are differentially treated based on their race and ethnicity.

Data availability

Data supporting the findings of this study are available from the corresponding author, upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ijdrr.2022.102855.

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